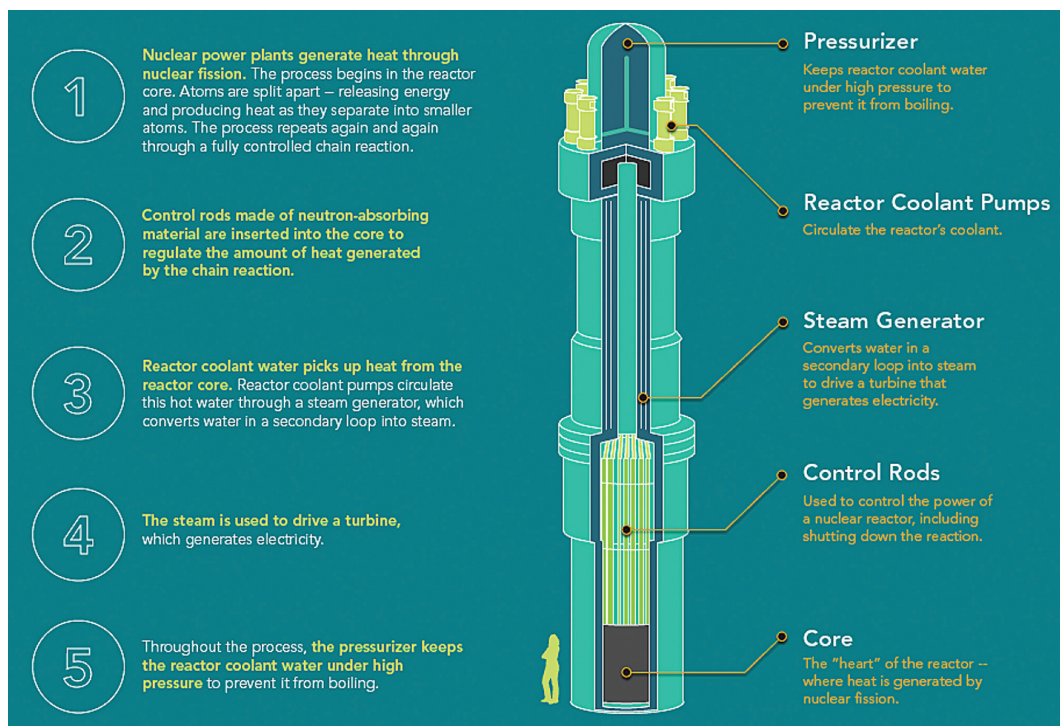




How a small modular reactor works (Source: www.energy.gov/na)

radiation exposure.

AI-driven design tools and digital twin technology are employed to simulate reactor performance. Rolls-Royce, for instance, is successfully using these tools for predictive maintenance and operational optimisation.

Designs vary in coolant types, fuel compositions, degrees of modularisation, technology readiness levels, and licensing readiness levels. This diversity presents challenges in selecting a design that suits specific energy demands, regulatory environments, and economic conditions. Key considerations

include radioactive inventory, nuclear safety, transport and handling constraints, and feasibility of fabrication and serial production.

The extent of modularity is influenced by factors such as cost optimisation, site conditions, construction complexity, build time, and overall project risk. A lack of operational data adds further uncertainty to the decision-making process.

Digital systems: The electronic core of SMR architecture

SMR designs are centred on digital electronic systems that serve as key enablers of scalability, safety,

and operational flexibility. These systems act as operational nerve centres, ensuring real-time monitoring, autonomous operation, and consistent performance under normal and emergency conditions by collecting, processing and analysing data from an array of sensors.

Even designs incorporating passive safety features depend on digital networks to monitor safety-critical parameters and automatically trigger protective responses when

thresholds are exceeded. Instrumentation and control (I&C) systems, human-machine interface systems, cybersecurity infrastructure, and advanced digital platforms are essential for achieving modularity, standardisation, and remote operability.

I&C systems. I&C systems are fundamental to SMR operation, continuously monitoring key reactor parameters and making real-time adjustments to maintain stability and safety. The I&C architecture comprises networks of sensors, actuators, logic controllers, and human-machine interfaces (HMIs).

These systems scan sensors to detect temperature changes, pressure variations, and shifts in neutron flux. The data is processed in real time to evaluate reactor status and execute automated control actions. If deviations are detected, the system responds by adjusting control rods, modulating coolant flow or initiating protective shutdowns.

Largely autonomous, I&C

Box 2: Human machine interface (HMI) technology

HMIs, integral to I&C systems, enable efficient human-machine interaction, offering intuitive platforms for monitoring and managing industrial processes. These include touch panels, graphical user interfaces (GUIs), and integrated displays that interface with programmable logic controllers (PLCs) and other industrial control systems to facilitate precise operations and control. Through HMI displays, operators manage both routine and complex tasks while gaining real-time insights into system performance. These systems also support grid management and energy distribution, ensuring stable and reliable power flow.